

Mayur Karmakar

Full-Stack Engineer | Applied AI Applications | React, Next.js & FastAPI
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SUMMARY

Full-Stack Engineer with 4+ years of experience building and deploying web applications across regulatory compliance, EdTech, and healthcare. Delivered AI-enabled products using React, Next.js, FastAPI, PostgreSQL, Azure OpenAI, and Azure Content Understanding, including structured LLM pipelines, Azure AI Search-backed RAG, OCR workflows, analytics, and source-level traceability.

TECHNICAL SKILLS

Frontend: React, Next.js, Redux, Zustand, TanStack Query, Tailwind CSS

Backend: Python, FastAPI, Node.js, Express.js, tRPC, Drizzle ORM, Prisma ORM, REST APIs

Architecture & Delivery: Micro-frontends, monorepo management, component-driven development, GitHub Actions CI/CD, Azure App Service, Azure Static Web Apps, Docker

AI & Document Processing: Azure OpenAI, Azure Content Understanding, OCR, Azure AI Search (document chunking, indexing, vector retrieval for RAG)

Integrations: Stripe, PayPal, Zoom, OpenVidu, Jitsi, Socket.IO, Clerk

DevOps & Tools: Git, Vercel, Azure Static Web Apps, GitHub Actions CI/CD, Supabase, Neon.tech, Turso

Databases & Platforms: PostgreSQL, Supabase, Neon, Turso

PROFESSIONAL EXPERIENCE

Full Stack Engineer - zipp.ai

Oct 2025 - May 2026

- Designed and built AuditPulse, a CGMP regulatory intelligence application using Next.js, FastAPI, PostgreSQL, Playwright, and GPT-4.1 through Azure OpenAI; developed an FDA warning-letter pipeline for scraping, data preparation, structured LLM classification, severity scoring, persistence, and retries.
- Delivered regulatory analytics dashboards with multidimensional filters, heatmaps, severity benchmarking, and time-series trend analysis.
- Built an AI-assisted batch-record review workflow using Azure Content Understanding OCR to extract process parameters, detect missing fields, and score data integrity.
- Implemented document chunking, indexing, vector storage, and retrieval with Azure AI Search to power RAG answers to GxP questions from selected SOP and regulatory documents.
- Developed a PDF viewer with source bounding boxes, deviation workflows, and 21 CFR Part 11-aligned audit trails; owned frontend, backend, database, AI integrations, and Azure delivery, including Docker and GitHub Actions deployments.

UI Developer - Bsecure (formerly Examd)

Sept 2023 - Jul 2025

- Developed and deployed two EdTech applications using Next.js, React, Drizzle ORM, and Tailwind CSS for examination and proctoring workflows.
- Built an AI-powered quiz generator that transformed text and uploaded files into structured question banks using OpenAI GPT; implemented RBAC, exam scheduling, auto-submission, and PayPal and Stripe payments.
- Implemented live video/audio monitoring using WebRTC and OpenVidu, including screen recording, real-time violation alerts, AI-based face and object detection, and Canvas LMS integration.

React & React Native Developer - Wisteli

Sept 2021 - Aug 2023

- Developed LMS-compatible online-proctoring features and real-time student-proctor communication using WebRTC and Socket.IO.
- Contributed to a cross-platform healthcare application using React and React Native, building analytical dashboards and improving diagnosis preprocessing and clinical data visualization.

PROJECTS

Chatbot (AI-Powered Conversational App) - Built a conversational AI application using OpenAI GPT with model selection, temperature controls, and API-key configuration.

Cowinistry (COVID-19 Vaccine Slot Predictor) - Built a web application that analyzed historical vaccination-slot data to estimate availability trends and delivered Telegram alerts.

EDUCATION

Bachelor of Computer Applications (BCA) - Trident Academy of Creative Technology (TACT),
Bhubaneswar

Aug 2018 - Sept 2021